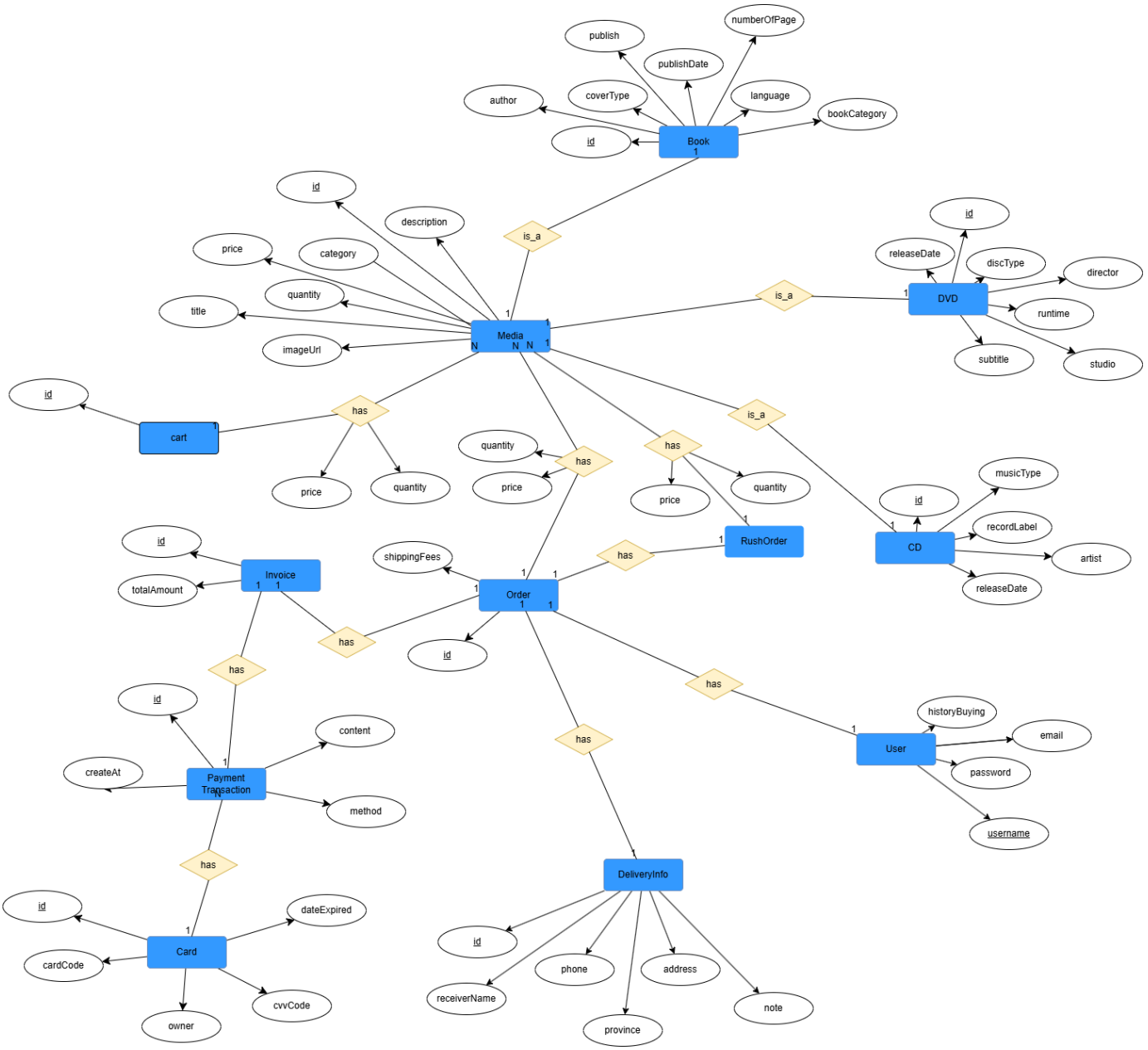
IT4490 – Software Design and Construction

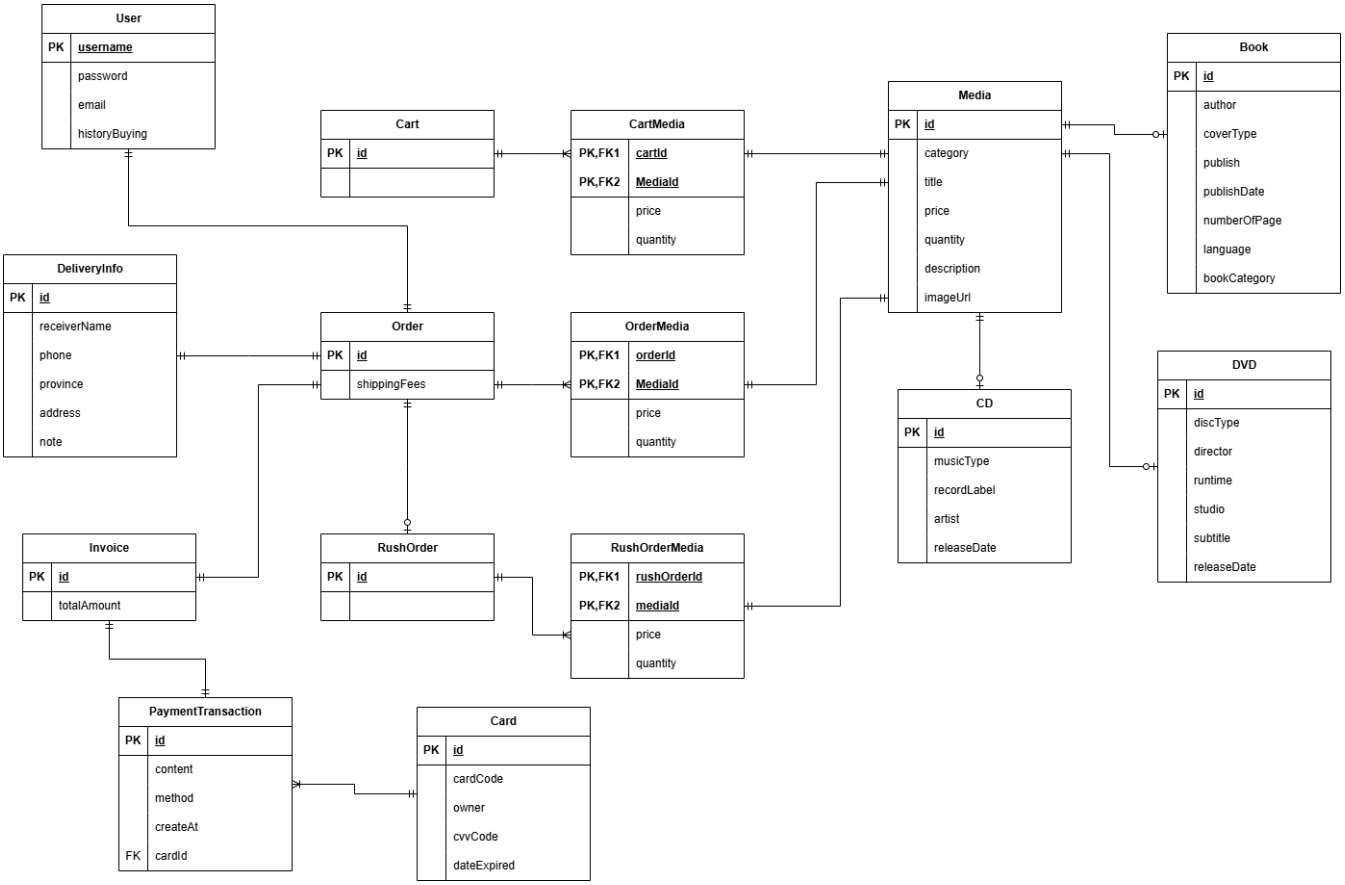
Nguyễn Hùng Cường – 20215264

Conceptual Data Model



Database Design

Logical Data Model



Physical Data Model

- **Media**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	x		id	Integer	Yes	ID, auto increment
2			category	VARCHAR(45)	Yes	Media type, e.g., CD, DVD
3			price	Interger	Yes	Current price
4			quantity	Integer	Yes	Number of products
5			Title	VARCHAR(45)	Yes	Product name
6			Value	Integer	Yes	Value of the product
7			imageUrl	VARCHAR(45)	Yes	Product image path

- **CD**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID, same as ID of Media of which type is CD
2			artist	VARCHAR(45)	Yes	Artist's name
3			recordLabel	VARCHAR(45)	Yes	Record label
4			musicType	VARCHAR(45)	Yes	Music genres
5			releasedDate	DATE	Yes	Release date

- **Book**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID, same as ID of Media of which type is Book
2			author	VARCHAR(45)	Yes	Author
3			coverType	VARCHAR(45)	Yes	Cover type
4			Publisher	VARCHAR(45)	Yes	Publishing house
5			publishDate	DATE	Yes	Date of publishing
6			numOfPages	Integer	Yes	Page number
7			Language	VARCHAR(45)	Yes	Language
8			bookCategory	VARCHAR(45)	Yes	Book category

- **DVD**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID, same as ID of Media of which type is DVD
2			discType	VARCHAR(45)	Yes	Disc type
3			director	VARCHAR(45)	Yes	Director
4			runtime	Integer	Yes	Duration
5			studio	VARCHAR(45)	Yes	Manufacturer
6			subtitle	VARCHAR(45)	Yes	Subtitles
7			releasedDate	DATETIME	Yes	Release date

#	PK	FK	Column Name	Data Type	Mandatory	Description
8			filmType	VARCHAR(45)	Yes	Genres

- **Cart**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	x		id	Integer	Yes	ID

- **CartMedia**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	cartId	Integer	Yes	ID
2		x	MediaId	Integer	Yes	Disc type
3			price	Integer	Yes	Total price
4			quantity	Integer	Yes	Number of products in cart

- **User**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	x		username	VARCHAR(45)	Yes	Username
2			password	VARCHAR(45)	Yes	Password
3			email	VARCHAR(45)	Yes	Total price

- **Order**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	x		id	Integer	Yes	ID
2			shippingFees	Integer	Yes	Shipping fees including common order and rush order.

- **OrderMedia**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	orderId	Integer	Yes	ID
2		x	mediaId	Integer	Yes	Disc type

#	PK	FK	Column Name	Data Type	Mandatory	Description
3			price	Integer	Yes	Total price
4			quantity	Integer	Yes	Number of products in order

- **DeliveryInfo**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID
2			receiverName	Integer	Yes	Shipping fees including common order and rush order.
3			phone	VARCHAR(20)	Yes	Phone Number
4			province	VARCHAR(20)	Yes	Province or City
5			address	VARCHAR(50)	Yes	Address
6			note	VARCHAR(100)	No	Note for shipper

- **RushOrder**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID

- **RushOrderMedia**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	rushOrderId	Integer	Yes	ID
2		x	mediaId	Integer	Yes	Disc type
3			price	Integer	Yes	Total price
4			quantity	Integer	Yes	Number of products in order

- **Invoice**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID
2			totalAmount	Integer	Yes	Total amount of products you buy

- **PaymentTransaction**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1		x	id	Integer	Yes	ID
2			content	VARCHAR(20)	Yes	Transaction contents
3			method	VARCHAR(20)	Yes	Payment methods
4			createAt	VARCHAR(20)	Yes	Date of creation
5		x	cardId	Integer	Yes	ID of used card

- **Card**

#	PK	FK	Column Name	Data Type	Mandatory	Description
1	x		id	Integer	Yes	ID, auto increment
2			cardCode	VARCHAR(45)	Yes	Card code
3			owner	VARCHAR(45)	Yes	Cardholders
4			cvvCode	VARCHAR(3)	Yes	CVV code
5			dateExpired	VARCHAR(4)	Yes	Expiration date